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Insights from Palm OS: The Future of the Mobile Computing Industry

Host: Coot Giles
Director Partner Marketing, PalmSource, Inc.



Driving Forward

We Are the Market Leaders in Enterprise

We Provide a Platform that Delivers the Right Solution



Panel Members

- Michael Rosenthal
  Manager, Strategic Alliances
  Handspring, Inc.
- Gary Koerper
  Director, Convergent Devices Business Segment
  Kyocera Wireless Corp.
- Andrew Breen
  Sr. Business Development Manager
  Palm Enterprise Strategy
- Doug Lloyd
  Director of Product Marketing
  Symbol Technologies, Inc.

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Handspring in the Enterprise

Mike Rosenthal
Manager, Strategic Alliances



What is Handspring doing in the enterprise?

Strong channel strategy with distribution through

- Ingram, Synnex, others
- Over 1,500 corporate resellers
- Wireless carriers

GSA schedule approved

Partnerships and support

- Solution providers
- Consultants
- System integrators

The Handspring Visor Family



Visor Deluxe

8MB
Expandable
Colors



Visor Neo

Very fast
New colors
Expandable
8MB



Visor Platinum

Very fast
8MB
Expandable



Visor Edge

Thin form factor
Detachable expansion
Rechargeable battery
8MB



Visor Pro

16MB
Rechargeable battery
Very fast
Expandable



Visor Prism

Photo quality color
Expansion
Rechargeable battery
8MB

Complete line of handheld computers

Springboard makes Visor a versatile expandable handheld for enterprises

Digital Camera



Content & Adapters



Wireless

Xircom
an Intel company



Presenter to Go



GPS



MAGELLAN

Bar Code Scanner



symbol

Sprint

- Data Backup & Memory Expansion
- Bar Code Readers
- Cellular Telephones
- Content Modules
- Digital Cameras
- Field Data Collection Probes
- Global Positioning System Modules
- Wireless LAN & WAN Modems

Over 70 modules

- Anywhere Email
- Corporate Data Synchronization (Exchange, Domino)
- Inventory Tracking
- Asset/Facilities Management
- Internet/Intranet
- Reference content (Medical, Auto Repair)
- Memory Expansion

The Handspring Treo





All your communications needs

Mobile phone, Palm OS® organizer, browser & messaging … in one small device

Redefining the applications



Phonebook

Datebook

Web

Messaging











Why Palm OS?

- U/I is easy and intuitive – better user experience
- Optimized for communicator sized products
- Large development community - 11,000+ Palm OS applications
- Large installed base – companies are supporting it - 60% of the Enterprise market*
- Runs on 33mhz processors – fast enough for most apps
- Lots of development support and resources
- Microsoft docs, spreadsheets, etc

***Source: IDC research, 10/01. n=1154**



Handspring devices are being deployed in enterprises today

- Target implemented 700 Handspring Visors and Symbol bar code scanners to make inventory management and tracking more mobile and productive
- Goodrich Aerospace outfitted personnel who service airplanes to improve productivity
- The Department of Defense implemented 1250 Handspring Visors to more effectively manage data from recruiting in the field, and tie back to their central system
- The Navy recently deployed Palm OS® devices to nearly half the crew on the destroyer USS McFaul to give servicemen and offices instant access to a variety of critical applications
- US Surgical has outfitted their sales force to be able to mobily connect to their SFA application to be more productive as they move about hospitals
- GE Medical has implemented 850 Handspring Visors integrated with their central database to more effectively track and manage assets



The right solution involves many components

Enterprise Solution

Device and peripherals	**Horizontal productivity apps (ex. Browser, PIM, email)**	**Integration with desktop applications**
Vertical productivity apps (ex. Order taking)	**Customization & Integration**	**Remote connectivity (ex. wireless data plan)**
Security	**Central management tools**	**Integration with corporate data systems**

Handspring Middleware Alliance Partners

AvantGo

VISTO

infowave

Extended Systems®
Beyond Connected™

XCELLENET™

Synchrologic™

neomar™
When Wireless Means Business

wireless knowledge

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Vision and Goals for Enterprise

Gary Koerper, Director of Converged Devices
Kyocera Wireless Corp.

Piecing together the enterprise puzzle



Kyocera Smartphone

Solution Providers

Customer Problem/Need

Carriers

System Integrators



Enterprise solutions

- Linchpin is the relationship with customers
  - Understanding their needs
  - Understanding their processes and politics
- Customers are slow to implement…..
- …..and even slower to change again.

- The battle for the customer relationship grow more competitive

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Blue Volt
Demo: Wireless Work Orders

Don Spear and David Vedinear

Blue Volt Inc.



BlueVolt – Leading Electricians' Daily Connection to the Internet

- PDA resident electrical calculators
- Wireless Work Orders including time and materials tracking
- Desktop companion applications and portal for work at home
- 10x Return On Investment



BlueVolt Wireless Work Order

- Database acts as a “data-broker,” not a permanent “data-store”
  - Easy to implement
  - Minimizes change to existing contractor systems
- Globally unique identifiers and revisioning guarantee data integrity and security
- The power of XML communication
  - Interoperability
  - Extensibility



Kyocera Smartphone Advantages

- Reduces hardware and communication costs with the combined benefits of pagers, cell phones, two-way radios and electrical calculators
- A handheld general computer that is easily connected to the Internet
- The cell phone is a familiar form factor that provides easy adoption by electricians



Enterprise Goals

- Work with solution providers in solving their customer needs/wants
- Support engagements and implementations from a hardware platform perspective
- Establish distribution and fulfillment process for the platform



Palm Enterprise Strategy

Andrew Breen, Sr. Business Development Manager
Former VP Business Development &
Infrastructure Products, ThinAirApps



The Mobile Situation

- Enterprises
  - Have more mobile workers than ever and need greater productivity
  - Have increasing security concerns around mobile information
  - Need an integrated, end-to-end solution for mobile email
  - Need to lower support and deployment costs
  - Fear manageability problems
- Developers
  - Need to deliver enterprise applications and data on the device
  - Want to extend existing business logic to the device through common standards
  - Want to leverage their existing skills and development tools
  - Want the difficulty of developing mobile solutions reduced
- Traditional Software Vendors
  - Need to extend their solutions to the device
  - Need a faster way to “mobilize” applications
  - Want their software to seamlessly run on all devices and networks

Palm Enterprise Software Strategy

Stage I: The Enterprise

- The Individual Solution: Desktop Link
- The IT Solution: Wireless Messaging Server
- The Benefits: Manageability, Security, and Flexibility

Target:

- **Individual Mobile Professional**
- **Corporate IT**

Stage II: The Developer

- The Materials: Palm Wireless Database Access Server
- The Tools: Third-party IDE integration
- The Future: More Mobile Enterprise Developer Products

Target:

- **Enterprise Developers**
- **Systems Integrators**

Stage III: The Reliable Transport for Mobile Applications

- The Infrastructure: A comprehensive platform for extending enterprise software
- The 5+1 Networks: Supports seamless deployment over all mobile networks: Bluetooth, 802.11, Mobitex, GPRS, CDMA, and the cradle

Target:

- **Strategic ISVs/OEMs**
- **Large Systems Integrators**
- **Skilled Enterprise Developers**

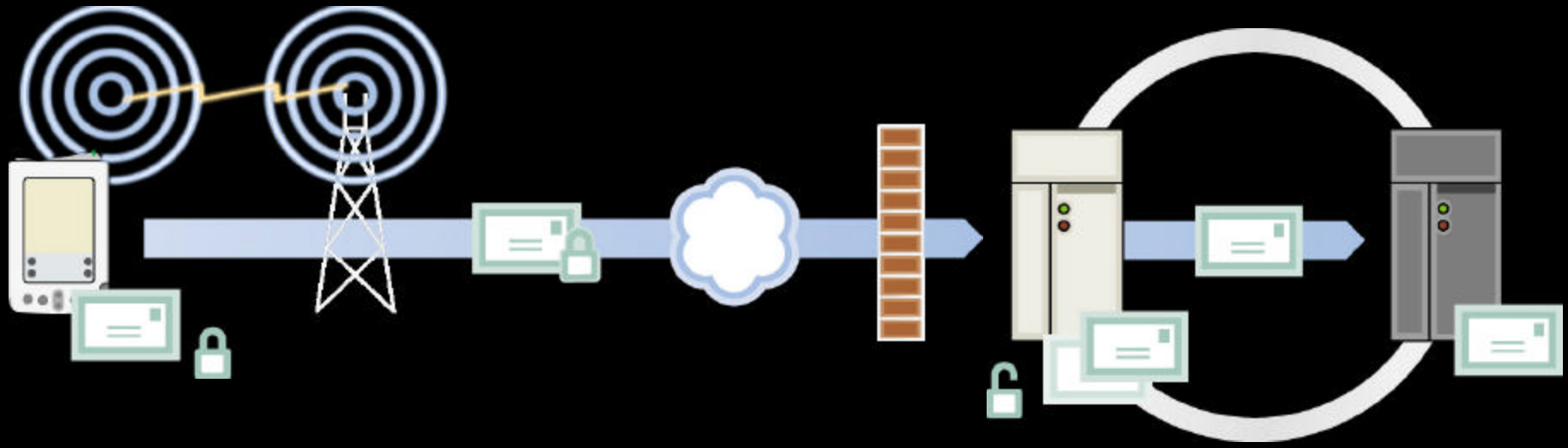


Palm Wireless Messaging Solution

An End-to-End, Palm-branded, Secure, Centrally-Managed Solution for Wireless “Always on” Enterprise Email

Palm Wireless Messaging Solution

Sending Email



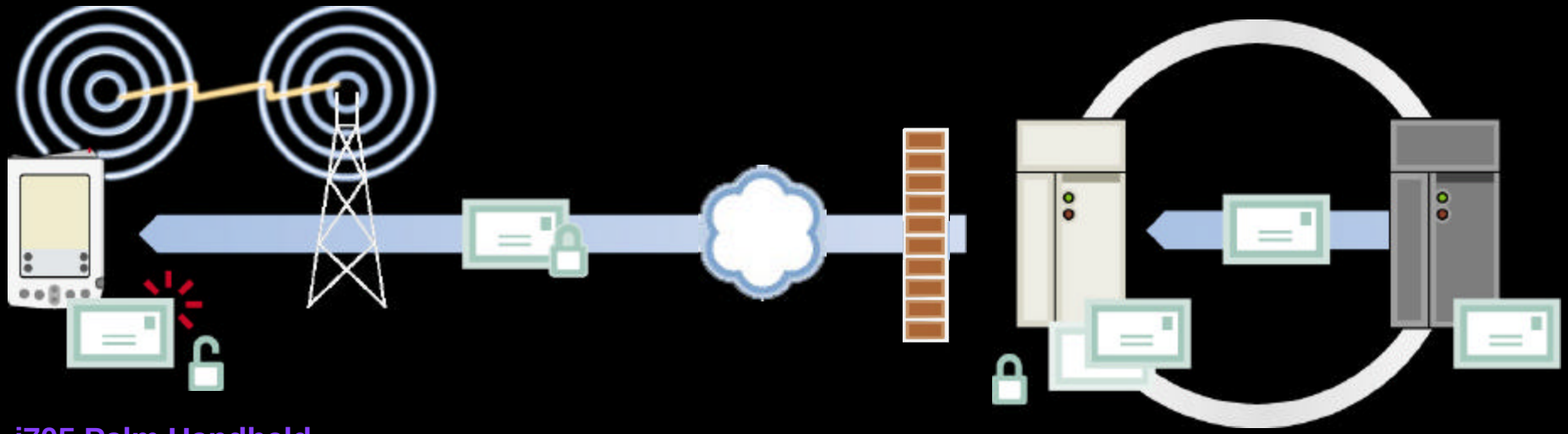
MultiMail Deluxe Email Client

Palm Wireless Messaging Server

Microsoft Exchange, or Lotus Domino

Palm Wireless Messaging Solution

Receiving Email



i705 Palm Handheld
MultiMail Deluxe Email Client

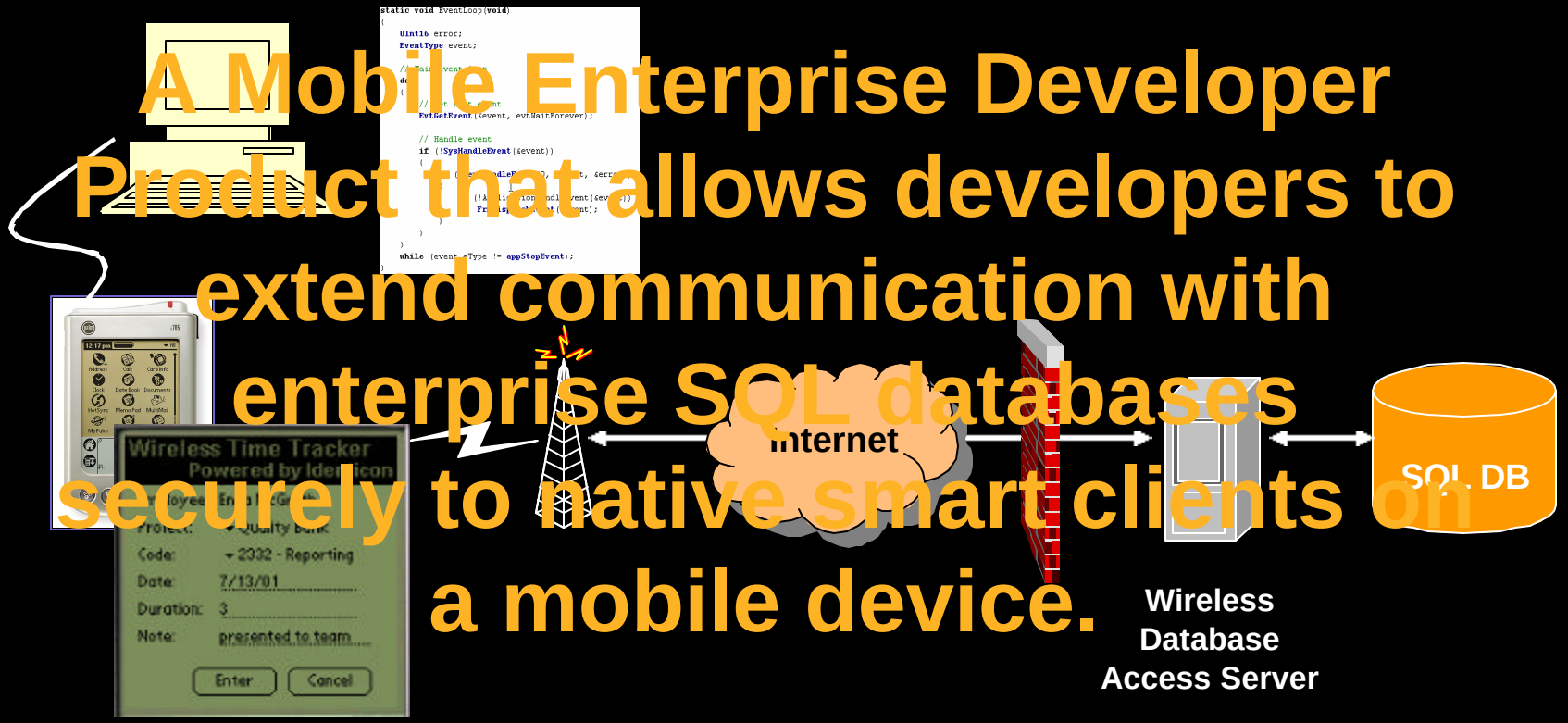
Palm Wireless Messaging Server

Microsoft Exchange, or Lotus Domino

WMS Key Features

- Supports Microsoft Exchange and Lotus Domino
- Secures email with End-to-End Encryption
- Pushes email out to user's Palm i705 handheld
- Utilizes Cingular Interactive (Mobitex) wireless network
- Runs on Windows NT and 2000 Server (UNIX coming)
- Is centrally managed and scalable
- Deployed behind corporate firewall

Wireless Database Access Server



A Mobile Enterprise Developer Product that allows developers to extend communication with enterprise SQL databases securely to native smart clients on a mobile device.



WDAS Key Features

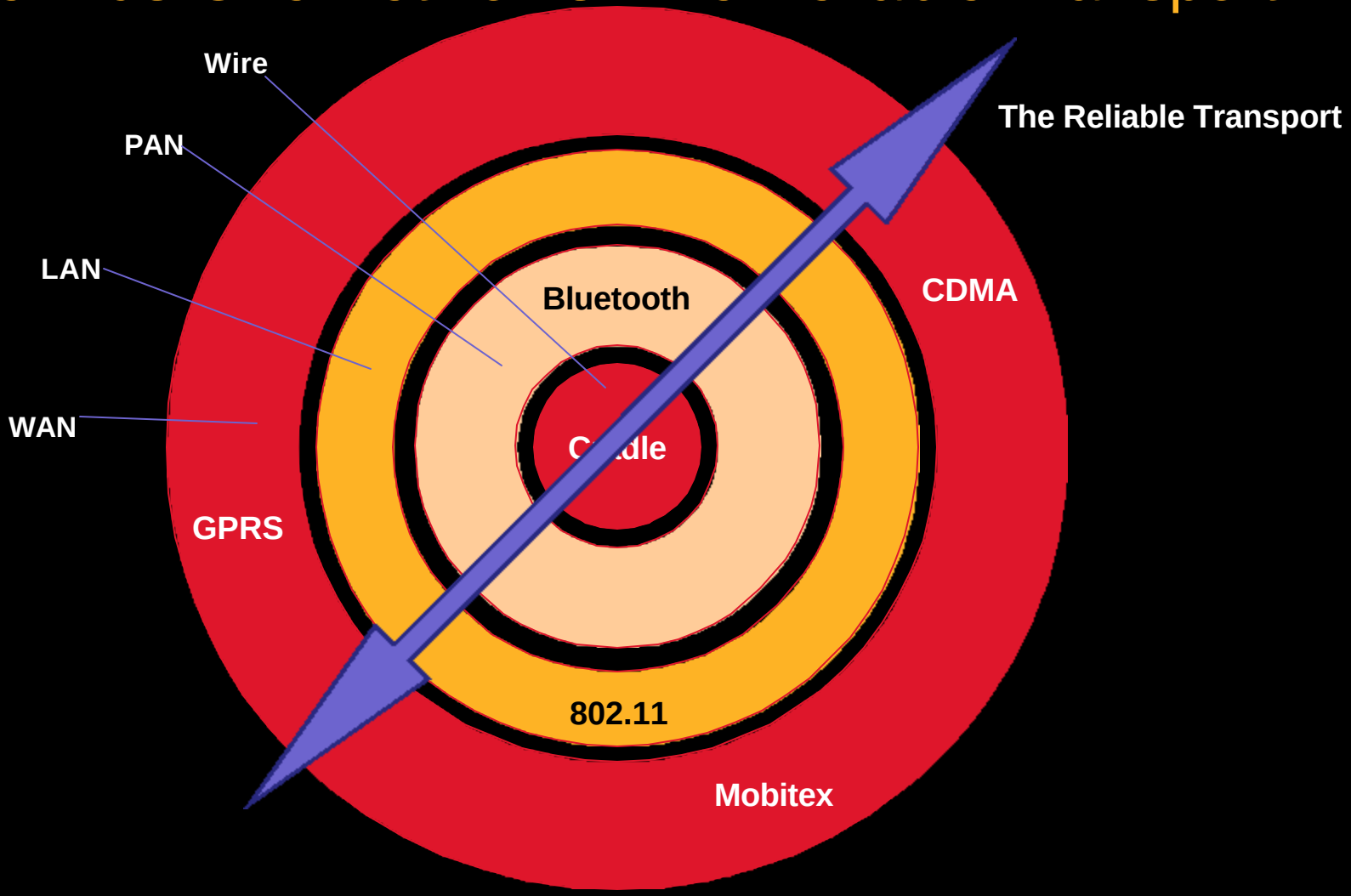
- Standards-based, integrated client/server environment
  - Allows developers to extend communication with enterprise SQL databases to mobile devices
  - The server sits behind-the-firewall and communicates via HTTP with End-to-End encryption
- Developers will be able to use familiar languages/IDEs
  - Developers build native smart clients in either Visual Basic with AppForge or C/C++ with Metrowerks
- Customization/Development is easier and faster
  - Developer has familiar environment and is masked from the complexity of wireless coding
  - Standard SQL communication leverages existing business logic already built into database
- First of the Palm Mobile Enterprise Developer Products



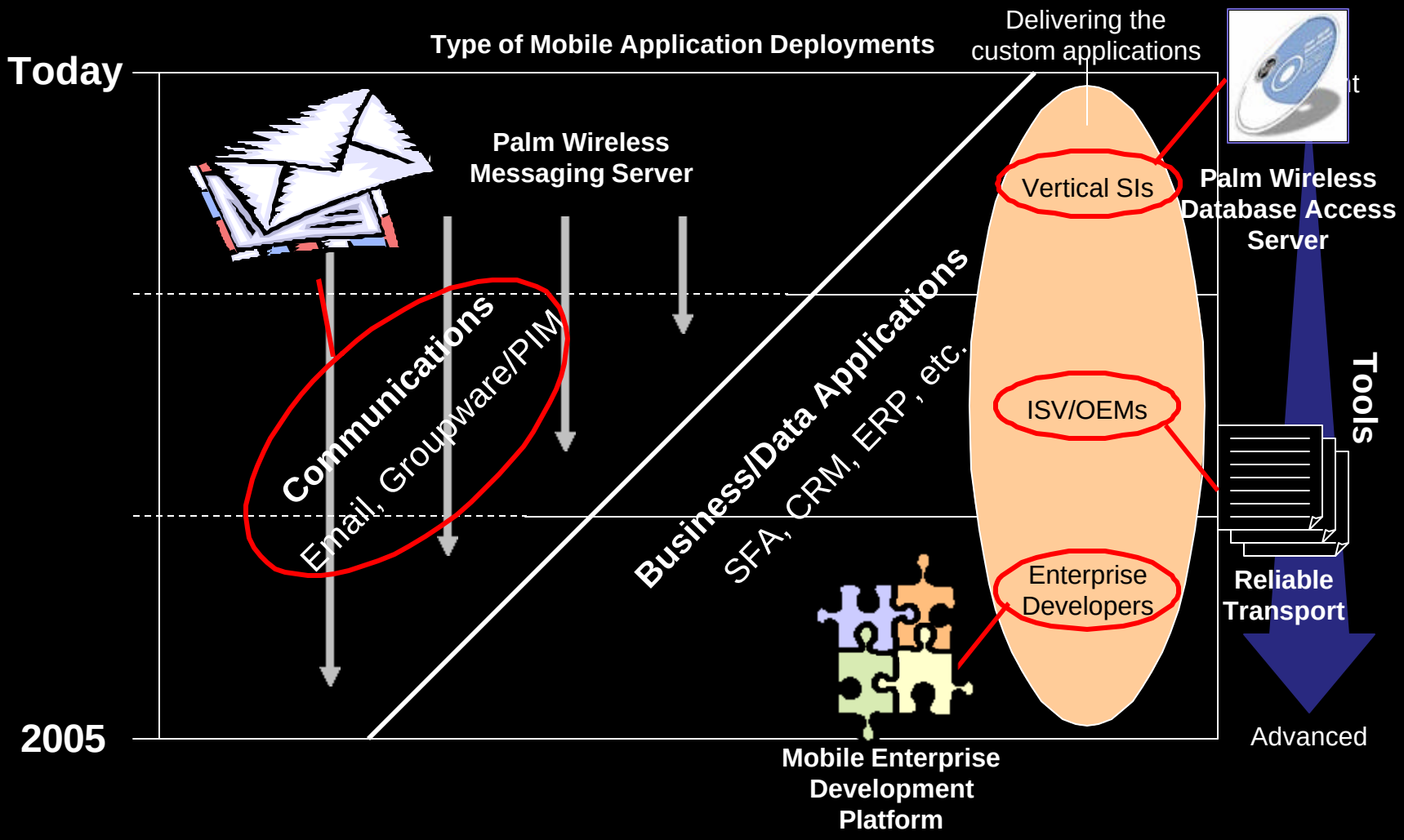
Mobile Enterprise Developer Products

- Databases
  - Wireless Database Access Server 1.1
- Remote Interface Systems
  - EJB/J2EE
  - Web Services: SunOne and Microsoft .Net
- Message Queues
  - JMS
  - MQ Series
- Browser Interfacing and Integration
  - New Palm Web Browser and plug-ins
- Common Languages and IDEs
  - Visual Basic via AppForge
  - C/C++ via Metrowerks
  - Java: VM Bundling

Five-Plus-One Networks: The Reliable Transport

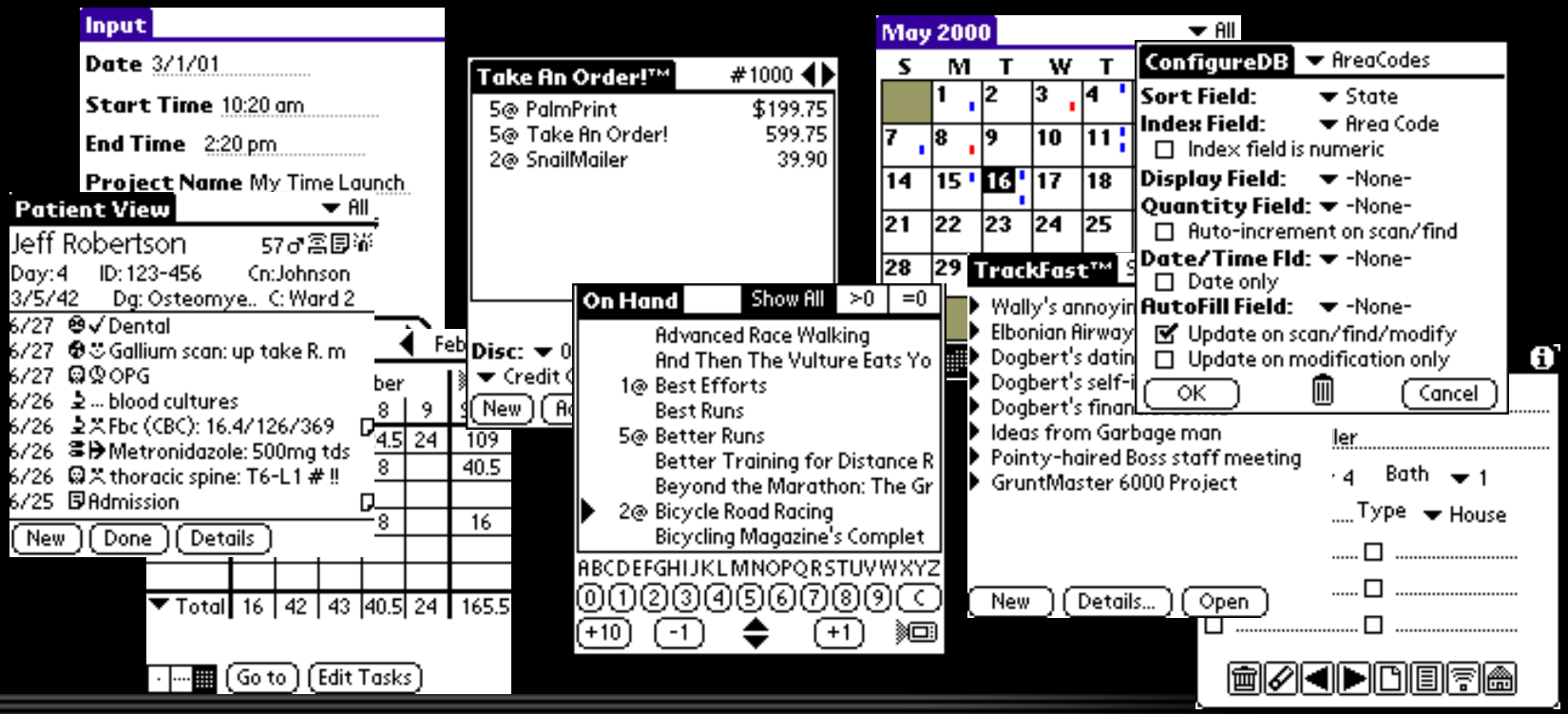


Mobile Enterprise Applications



Applications Developers Are Critical

- Over 190,000 developers (Adding 1,300 per week)
- Over 13,000 apps
  - Sales Force Automation and Customer Relationship Management
  - ERP and Field Service Automation



The State of Palm, Inc. in the Enterprise

- Palm owns over 60% of the overall handheld market
- Palm is already the leader in the enterprise
  - Over 80% of Palm handhelds sold today are used business purposes
- Expanding Enterprise handheld offering
  - Executive: Palm m500 series with more this year
  - Mobile Data: Palm i705 handheld and beyond
- Refocused direct and channel sales forces
- Acquired the software intellectual property and enterprise expertise of ThinAirApps
  - Products being delivered today
  - Infrastructure and Tools to support partners and developers
- An integrated offering from one vendor: devices, software, service, and support
  - What the enterprise wants and needs

ARM
THE ARCHITECTURE FOR THE DIGITAL WORLD™





802.11b

CSD GPRS

CDMA/1xRTT



Symbol Technologies, Inc.

Doug Lloyd
Director of Product Marketing

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Mobilizing the Enterprise

Douglas Lloyd
Director of Product Marketing
Symbol Technologies



Symbol Technologies Inc.

Core Competencies

Enterprise Mobility

Enterprise Applications

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Core Competencies

Scanning





3rd Party Products

Professional Services

Collaborative Solutions

RFID Integration

WWAN Integration

Platform Integration

Wireless Networks



Application-Specific Handheld Computers



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Mobility

or

Portability?

symbol



Enterprise Mobility

Wireless LANs are different, and will drive new mobility paradigms

- Change how we communicate
- Drive new business paradigms
- Drive new related technologies
- Will be as pervasive as the Internet

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Enterprise Mobility

- LANs provide fixed access with high data rates
- WLANs provide highly mobile access with lower data rates
- Applications will be different
- Requirements are different
- Management is different

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Enterprise Mobility

"The **mobile** information appliance market will enjoy exponential growth over the next few years, with annual worldwide sales exceeding 25 million units by the end of 2002," said David Hayden, Mobile Insights' new senior industry analyst focusing on mobile information appliances.

"Wireless communications will be the catalyst fueling the growth of this new mobile market, and as the 'killer application,' will lead to pervasive use of handheld computers."

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Enterprise Mobility

New Challenges

- Security
  - Untrusted vs. trusted users:
  - Protecting all of the network resources
- Management
  - Multiple devices, OS's, Drivers, Applications
- Resource Allocation
  - Bandwidth
  - QOS
- Mobility
  - Consistent Services
  - Persistent connectivity

Weaving the Mobile Enterprise Fabric™

Network Management
Mobile Management
Application Management
Security services



QOS – Voice & Data
Power Over Ethernet
Device Services
Radio Services

Pre-emptive Roaming
Load Balancing
Power Management
Security *with mobility*

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Enterprise Mobility

74% consider wireless applications strategic to their business

Source: eMobility Wireless Study, November 2000

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Enterprise Applications

Bar Code Medication Administration



Enterprise Applications – Mobilizing Healthcare



Patient Lookup

Scan Patient Wristband

Name: Keith D Kztoar

SSN: 408-02-9575

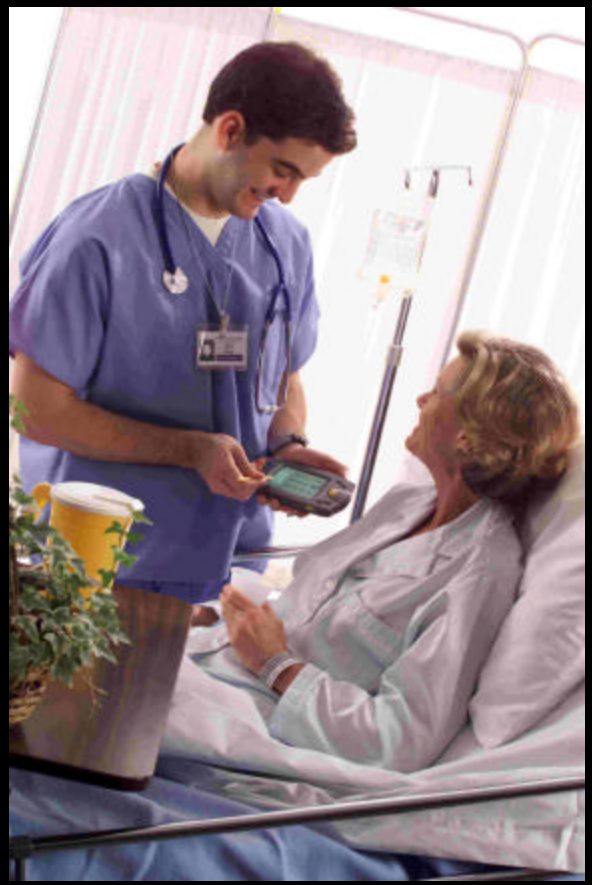
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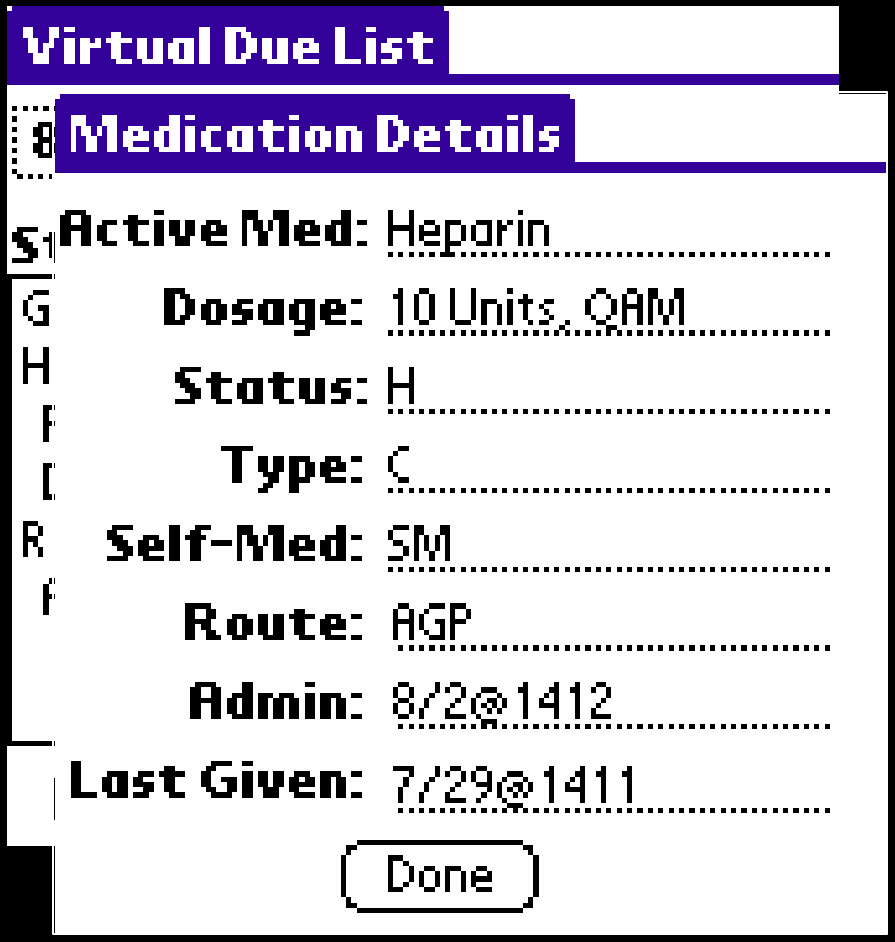
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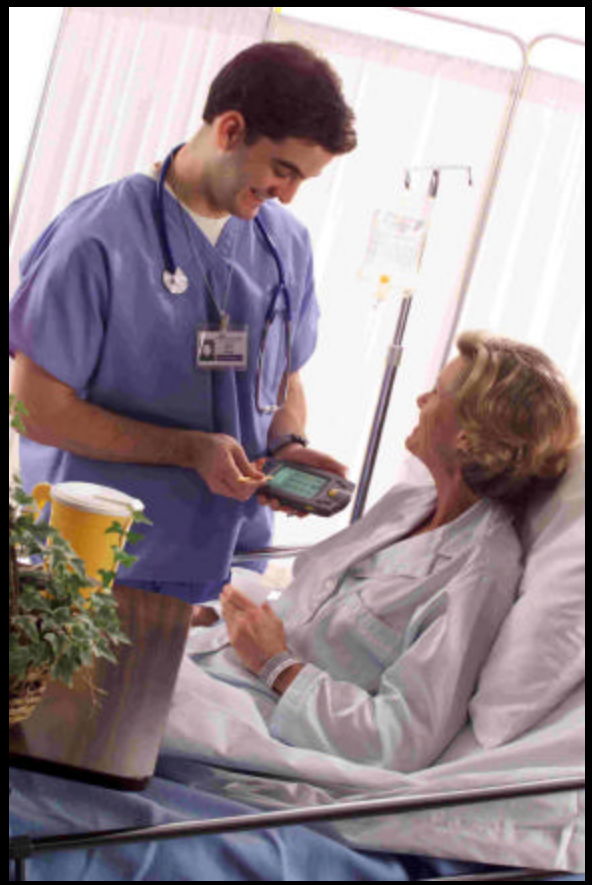
Enterprise Applications – Mobilizing Healthcare

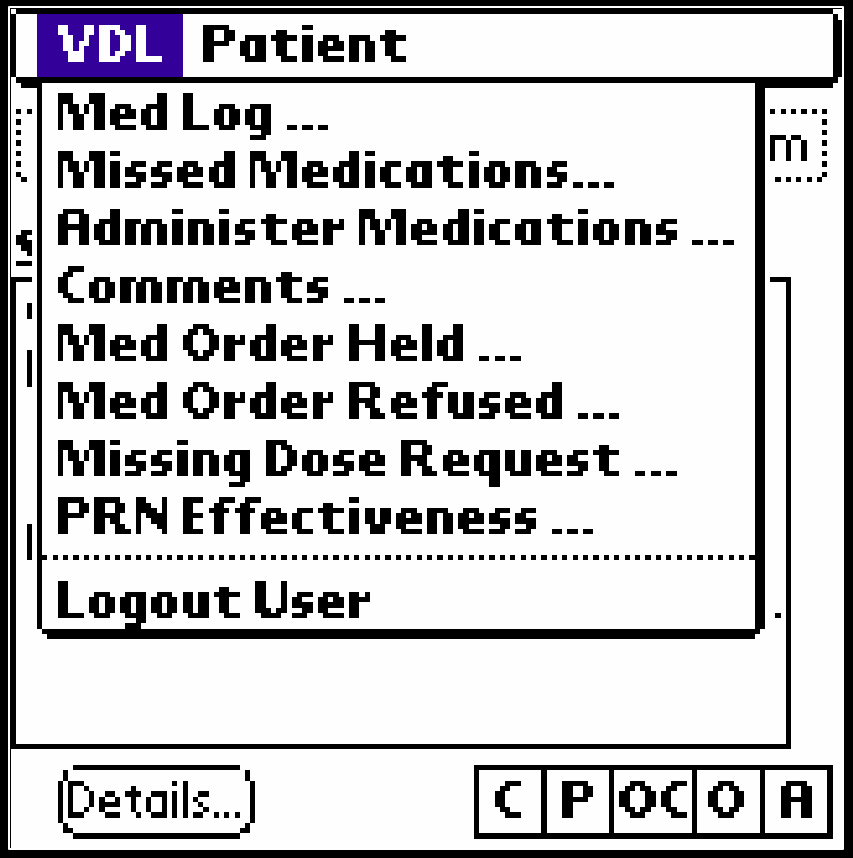




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Enterprise Applications – Mobilizing Healthcare





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Enterprise Applications – Mobilizing Healthcare

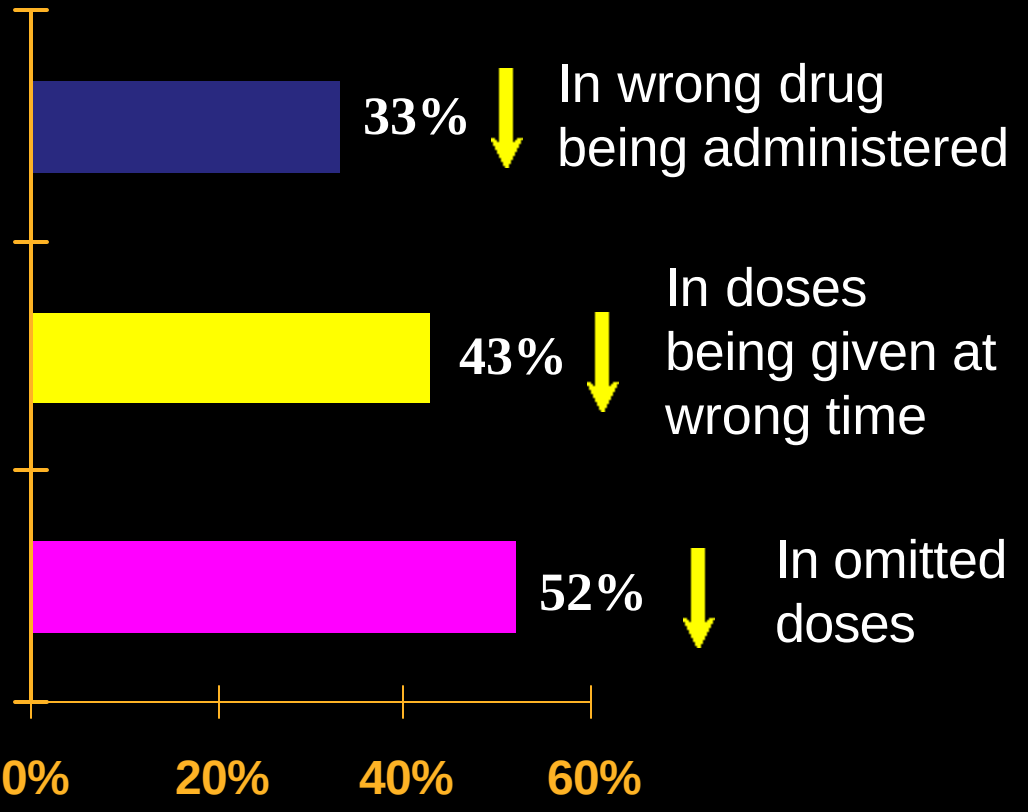




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Enterprise Applications

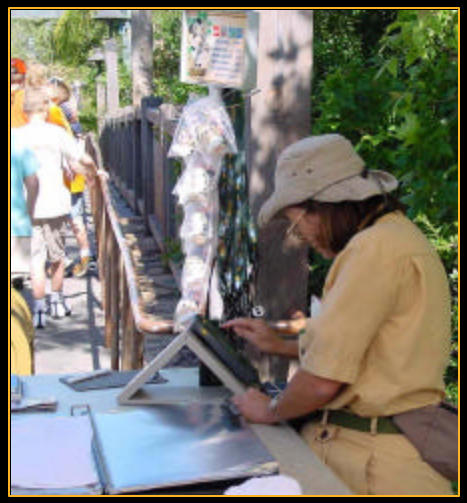
- Impact of Integrating Bar Coded Medications with a Point-of-Care System

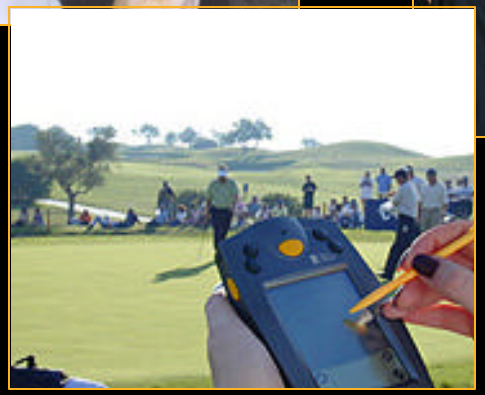


North Colorado Medical Center Study, AJHP

Enterprise Applications – A new Mobile world











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